DATA MINING LAB

Midterm Lab Exam

Disease: Malaria

Submitted by Group 9 :

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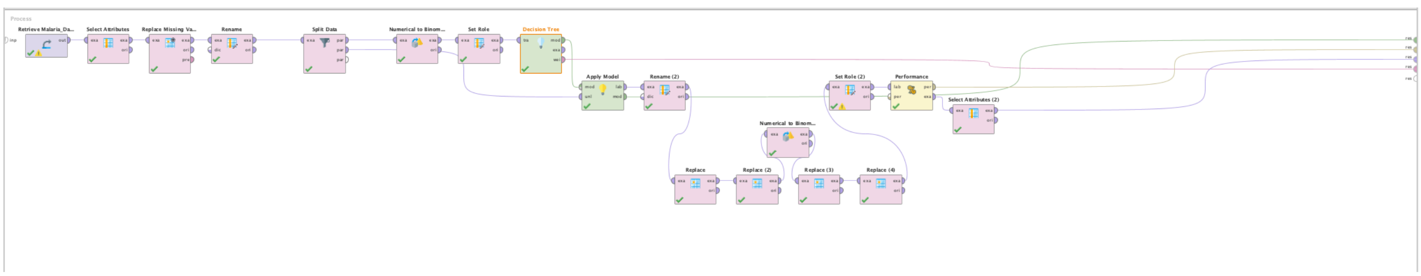
DATA DICTIONARY:

Column Name	Description	Data Type	Example Value	Remarks
Age	Represents the patient’s age in years.	Integer	52	Helps identify age-related malaria risk.
Fever	Indicates if the patient experienced fever (1 = Yes, 0 = No).	Integer (0/1)	0	Fever is a common symptom of malaria.
Headache	Indicates if the patient reported a headache (1 = Yes, 0 = No).	Integer (0/1)	0	Often occurs with malaria infection.
Abdominal_Pain	Indicates if the patient had abdominal pain (1 = Yes, 0 = No).	Integer (0/1)	0	Can be associated with severe malaria.
General_Body_Malaise	Indicates if the patient felt general body weakness or fatigue (1 = Yes, 0 = No).	Integer (0/1)	1	A common symptom during malaria infection.
Dizziness	Indicates if the patient experienced dizziness (1 = Yes, 0 = No).	Integer (0/1)	0	May occur due to fever or dehydration.
Vomiting	Indicates if the patient experienced vomiting (1 = Yes, 0 = No).	Integer (0/1)	0	Common in moderate to severe malaria cases.
Confusion	Indicates if the patient showed confusion or disorientation (1 = Yes, 0 = No).	Integer (0/1)	0	Seen in severe malaria or cerebral cases.
Backache	Indicates if the patient complained of back pain (1 = Yes, 0 = No).	Integer (0/1)	1	A possible symptom of infection or weakness.
Chest_Pain	Indicates if the patient reported chest pain (1 = Yes, 0 = No).	Integer (0/1)	0	May occur due to muscle fatigue or infection.

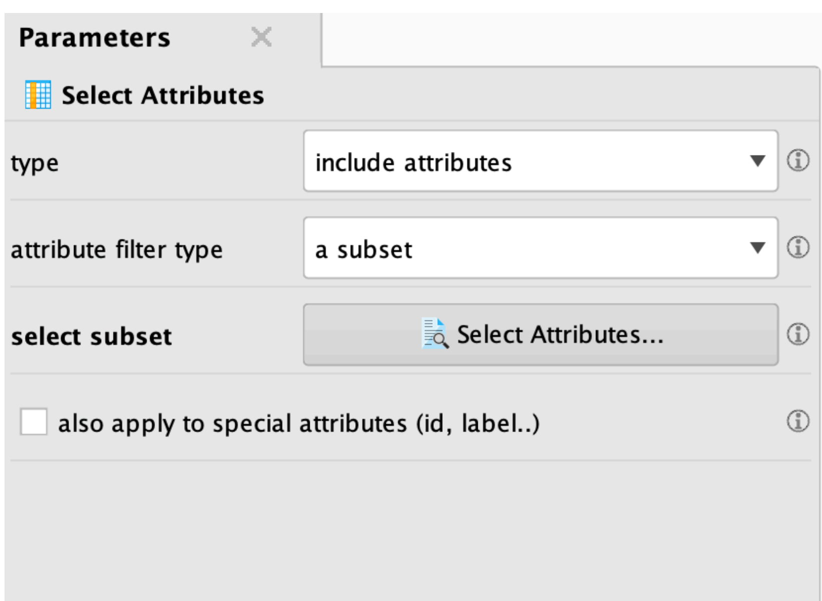
Coughing	**Indicates if the patient experienced coughing (1 = Yes, 0 = No).**	**Integer (0/1)**	**0**	**Not a core malaria symptom but may appear in co-infections.**
Joint_Pain	**Indicates if the patient had joint pain (1 = Yes, 0 = No).**	**Integer (0/1)**	**0**	**Can be a sign of body inflammation due to infection.**
Target	**Indicates malaria diagnosis (1 = Positive, 0 = Negative).**	**Integer (0/1)**	**0**	**Used as the prediction label in model training.**

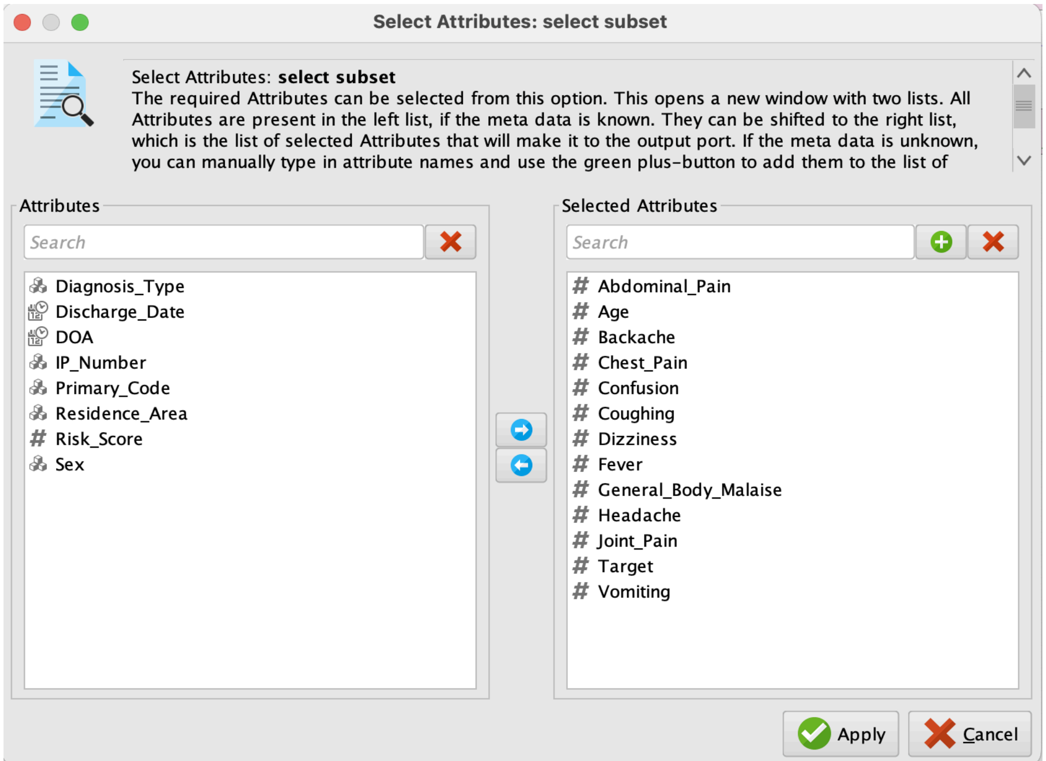
DESIGN:



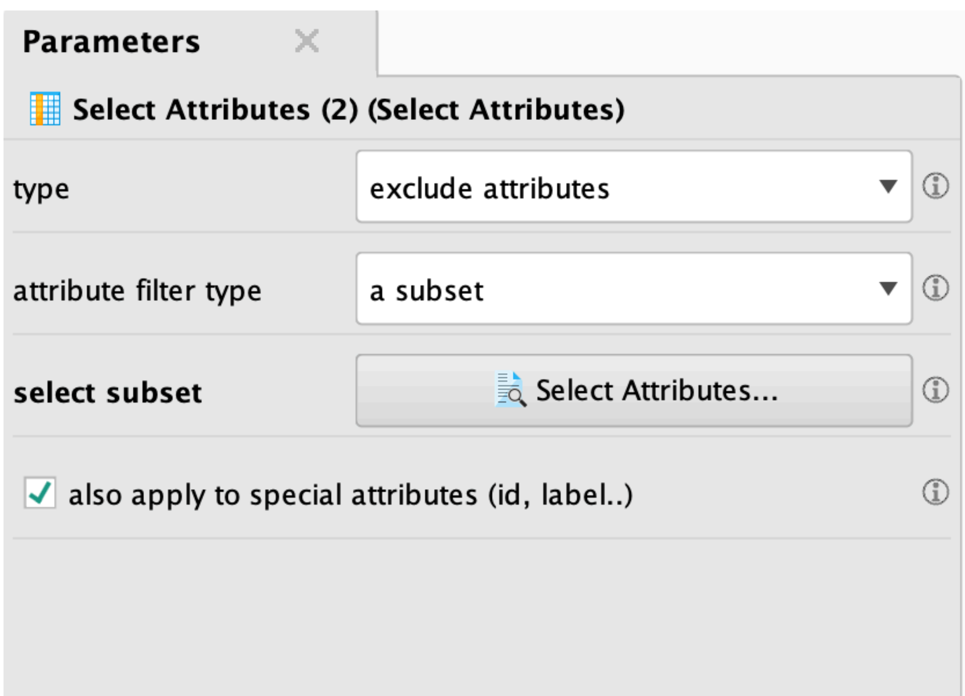
PROCESS DIAGRAM: *The process retrieves, cleans, and prepares the malaria dataset, applies a Decision Tree model to predict malaria risk, then renames and replaces output values for better readability before evaluating the model's performance.*

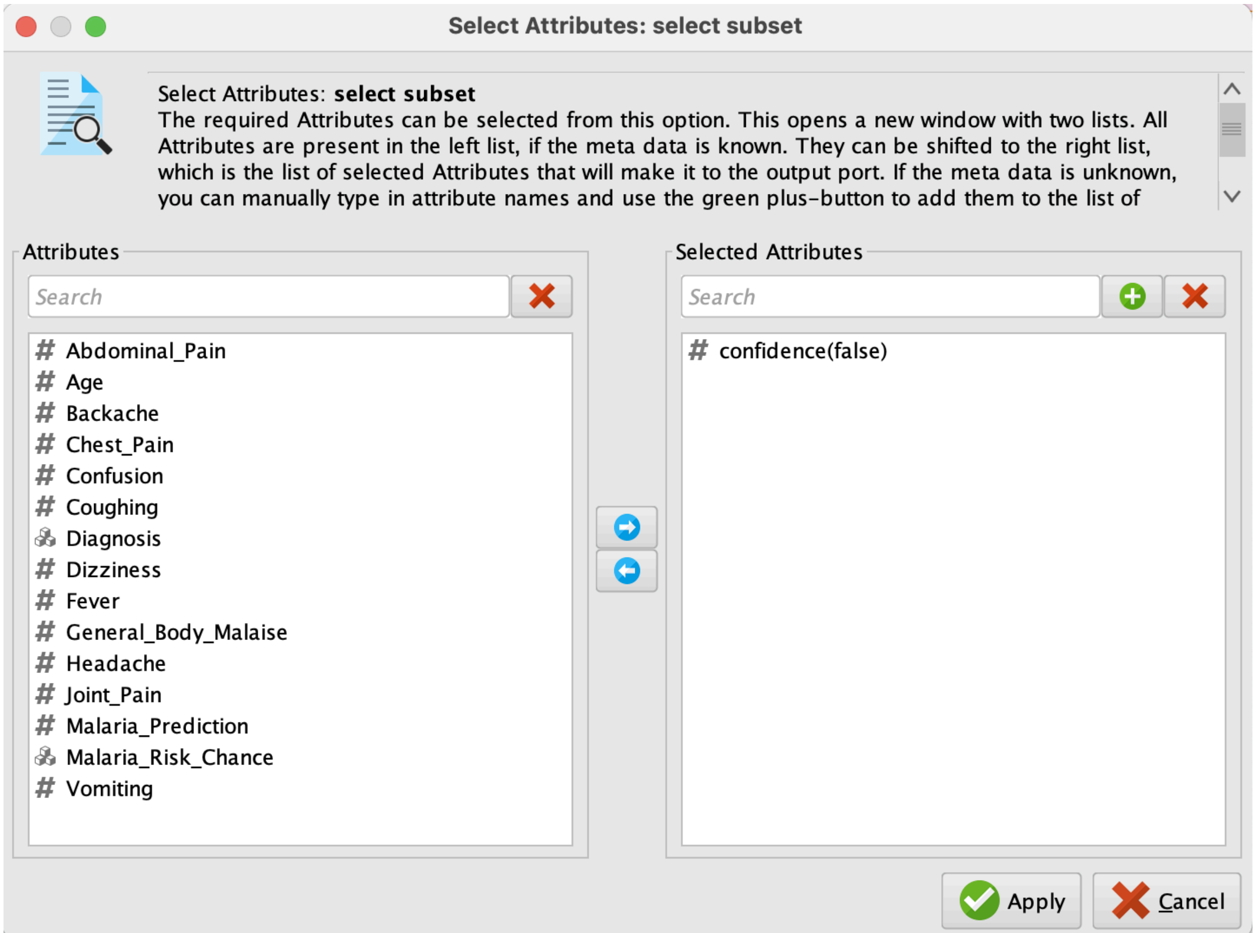
SELECT ATTRIBUTES:





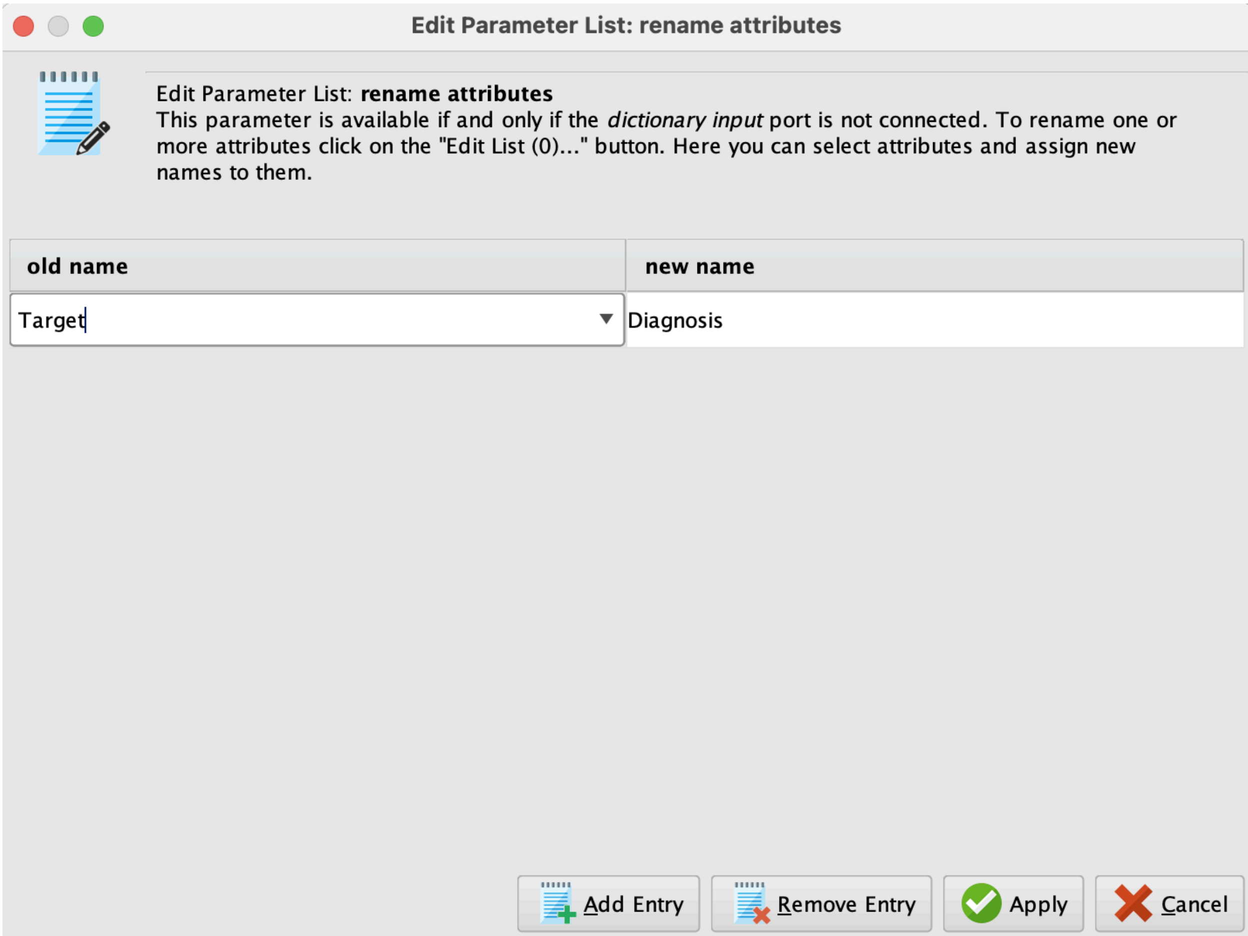
***SELECT ATTRIBUTES**: Used to select the important attributes that will be used for training and testing the model.*



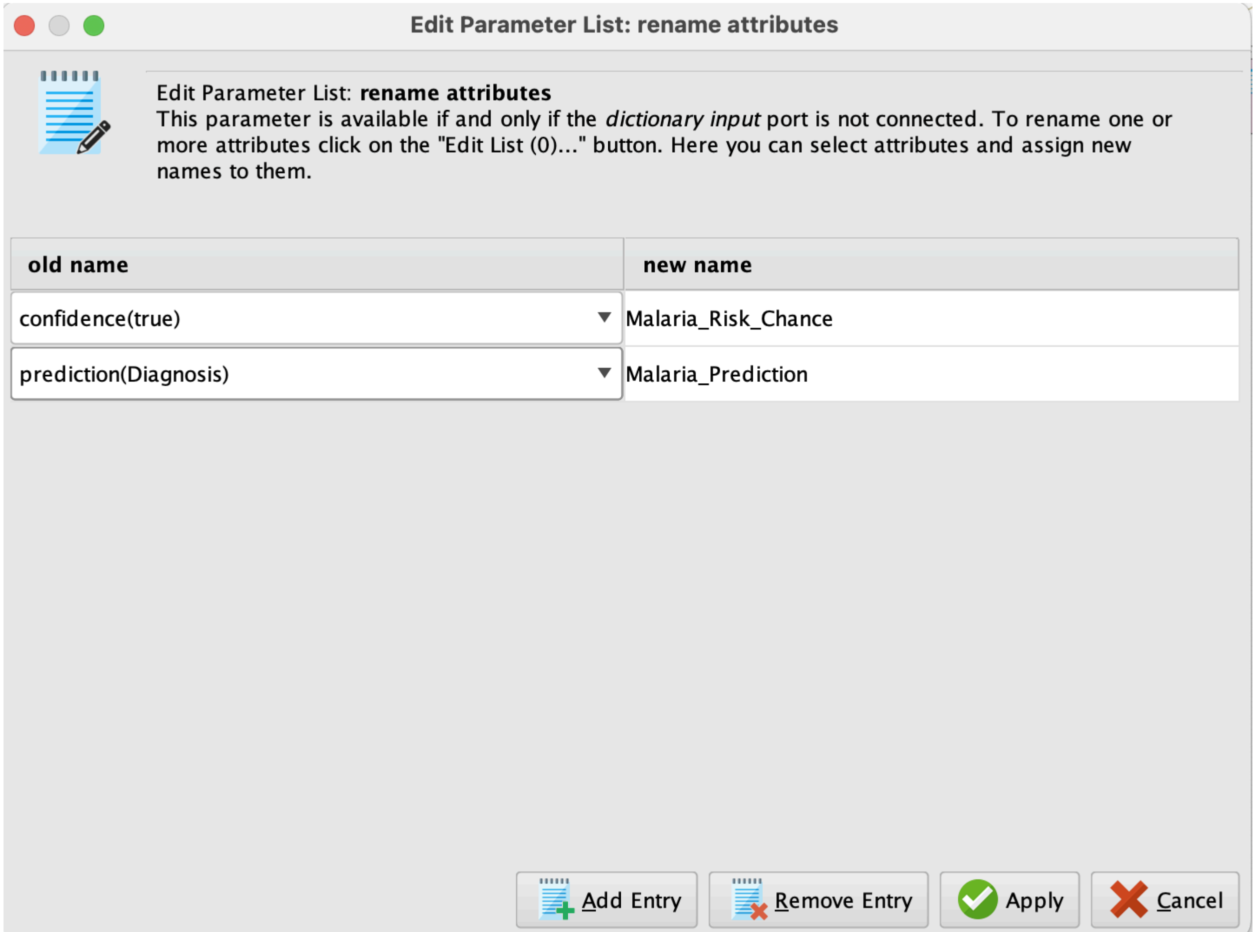


***SELECT ATTRIBUTES**: Used to exclude the confidence(false) column because it is not needed for the final results table of our Malaria data.*

RENAME ATTRIBUTES:

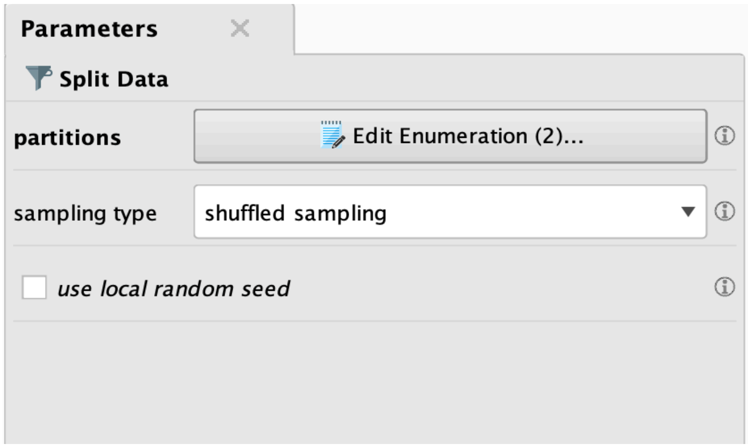


***Rename 1**: Renaming column title "Target" to "Diagnosis" for better data representation*

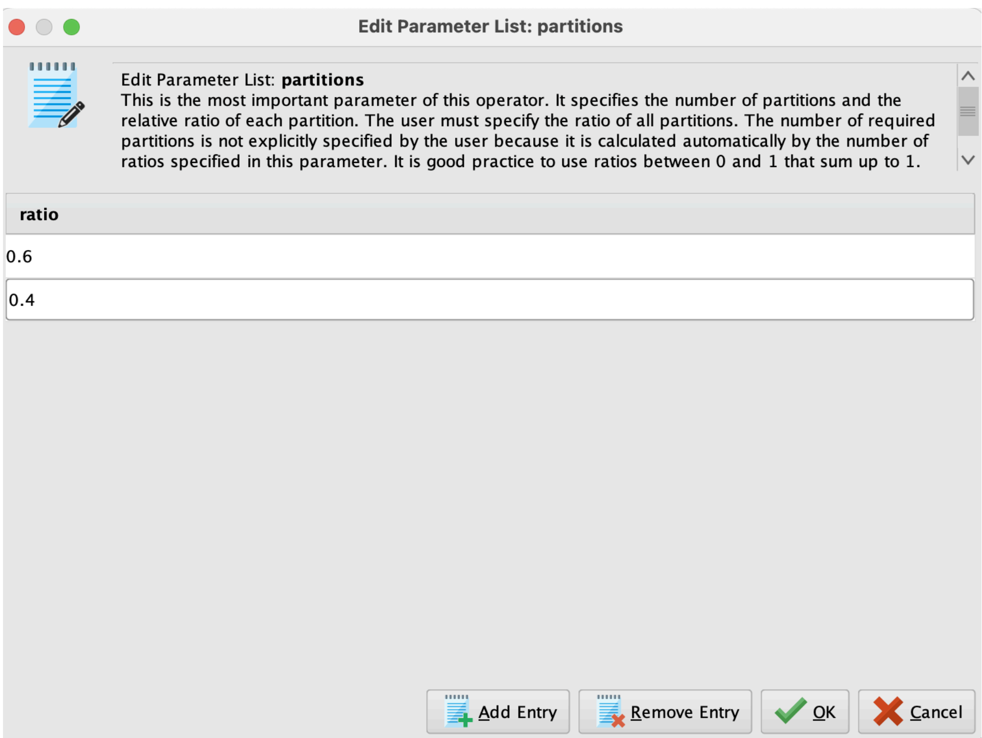


Rename 2*: Renaming column title "confidence(true)" (confidence of the model that the patient has malaria) to "Malaria_Risk_Chance" for better data representation and "prediction(Diagnosis)" (model's prediction if the patient has Malaria) to "Malaria_Prediction"*

SPLIT DATA:

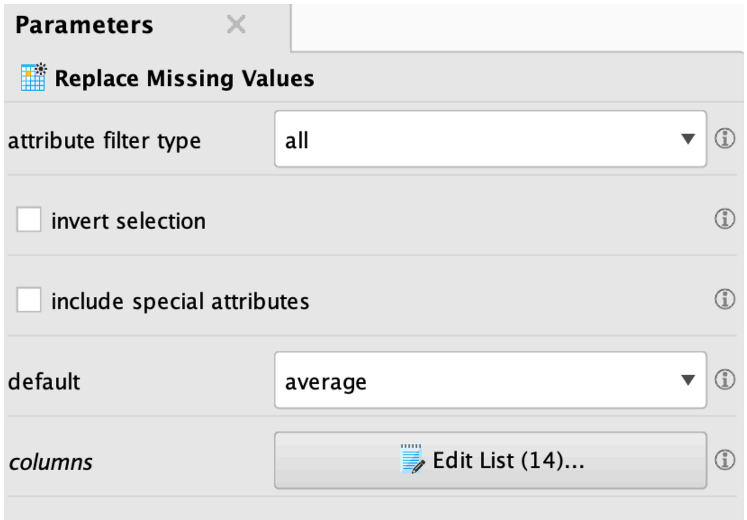


***Split Data:** Makes the training and testing sets have a fair mix of different kinds of cases by randomly mixing (shuffling) all the rows.*



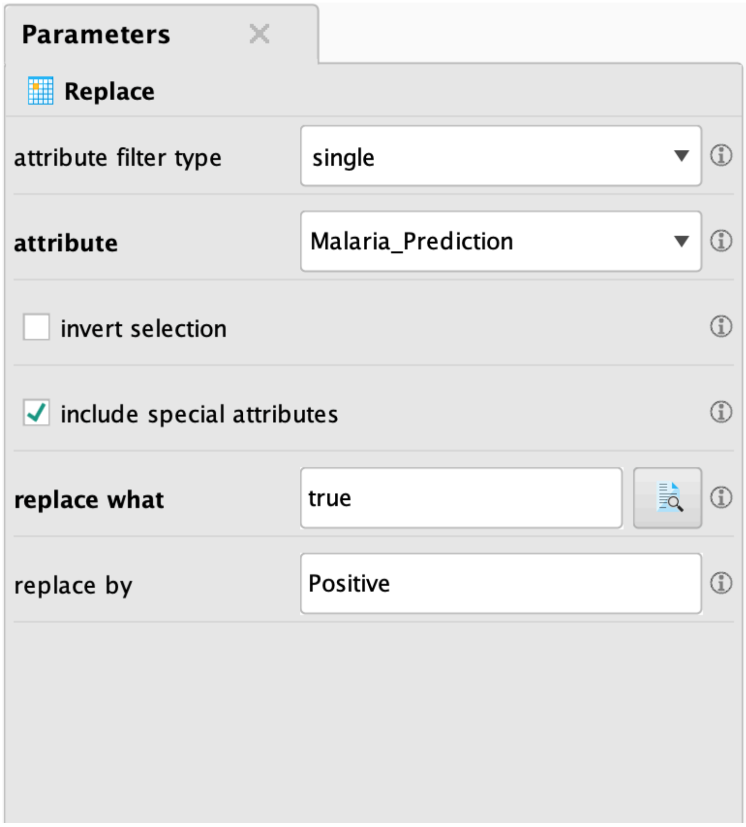
***Partitions:** 60-40 split was used to ensure the model learns from sufficient data while having enough reliable data for testing.*

REPLACE MISSING VALUES:

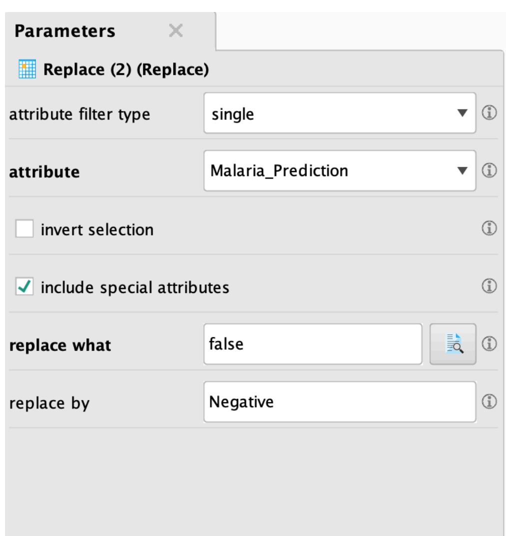


Edit Parameter List: *Replacing the missing values in the dataset to the most appropriate default value*

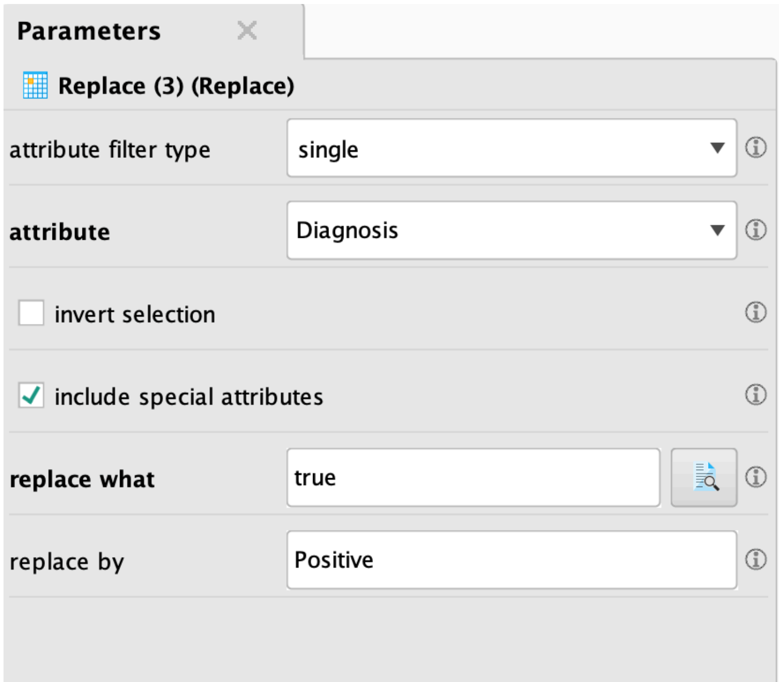
REPLACE:



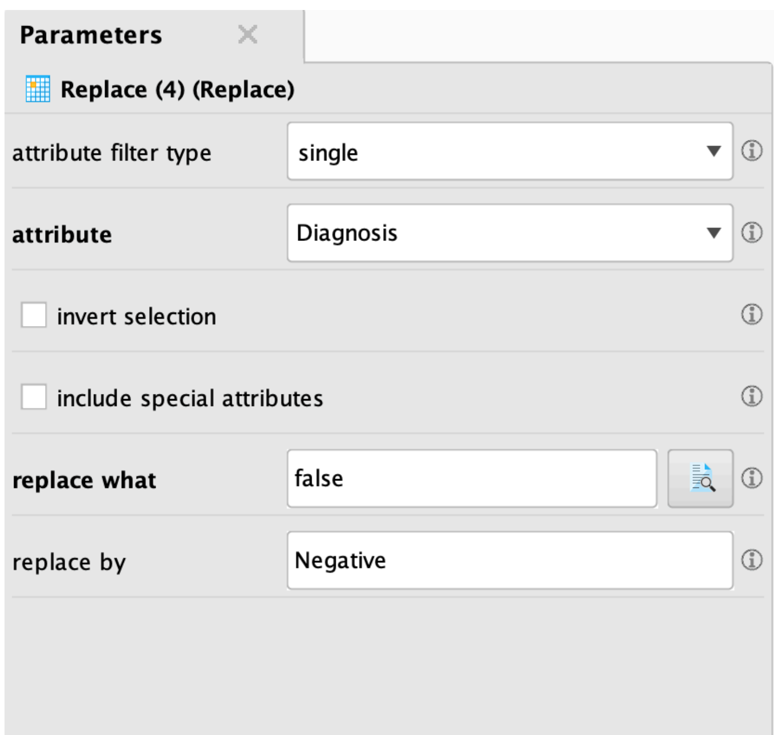
Replace 1: *Replacing all values labeled "true" to "Positive" in the Malaria_Prediction column for better data representation.*



Replace 2: *Replacing all values labeled "false" to "Negative" in Malaria_Prediction column for better data representation*

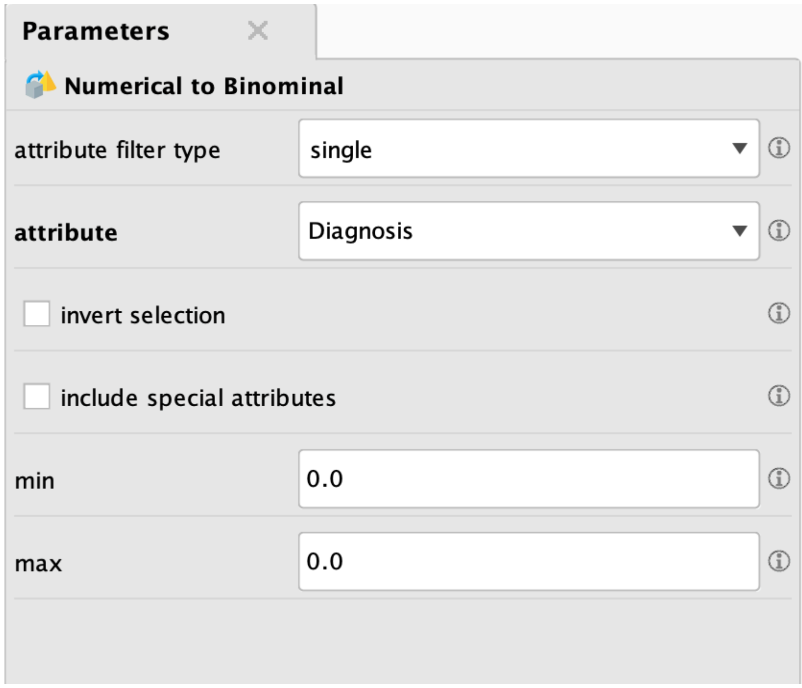


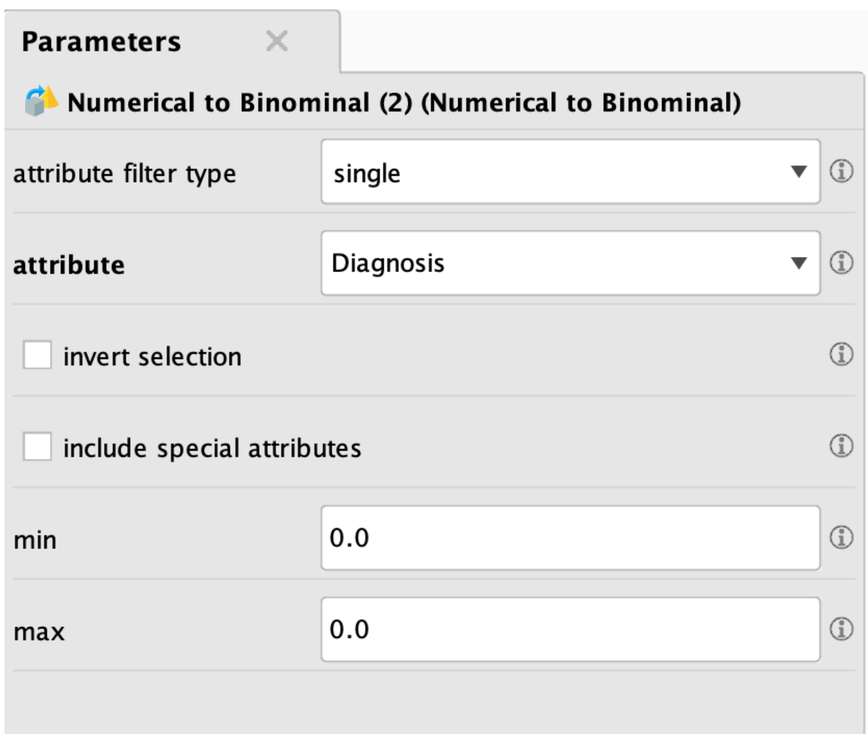
Replace 3: *Replacing all values labeled "true" to "Positive" in the Diagnosis column for better data representation.*



Replace 4: *Replacing all values labeled "false" to "Negative" in Diagnosis column for better data representation*

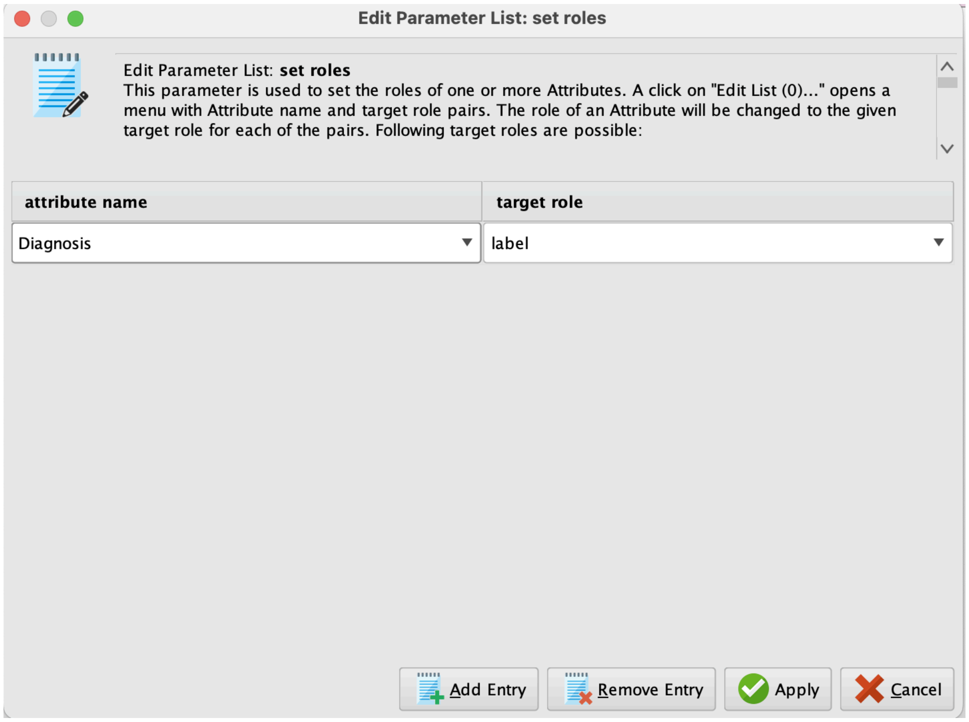
NUMERICAL TO BINOMIAL:

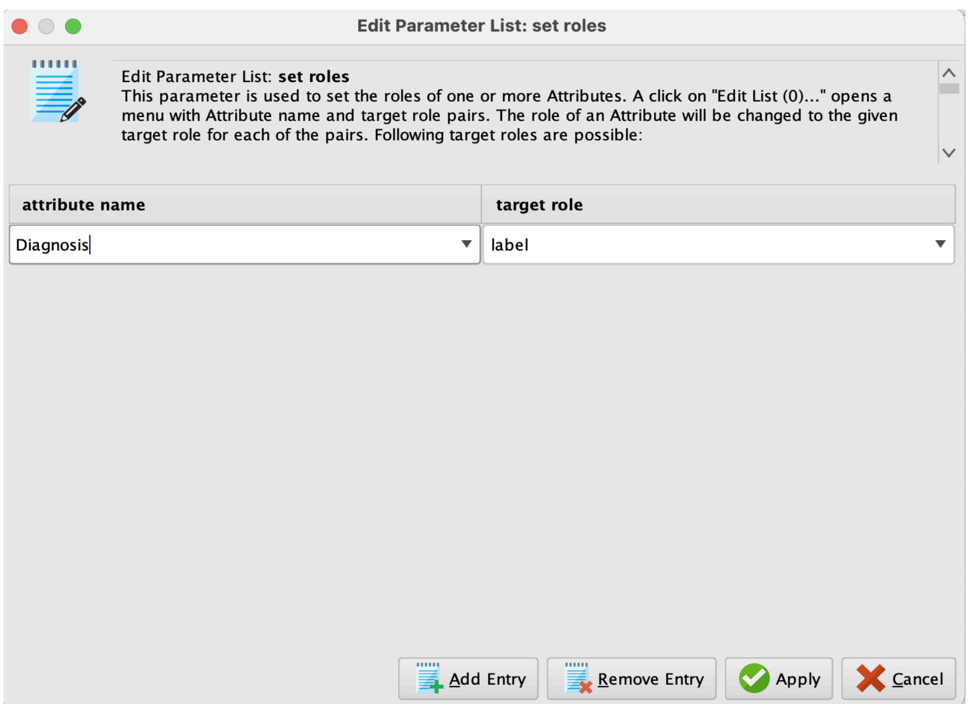




Parameter: *Used to convert numerical values into two categories so the model can treat them as classification labels*

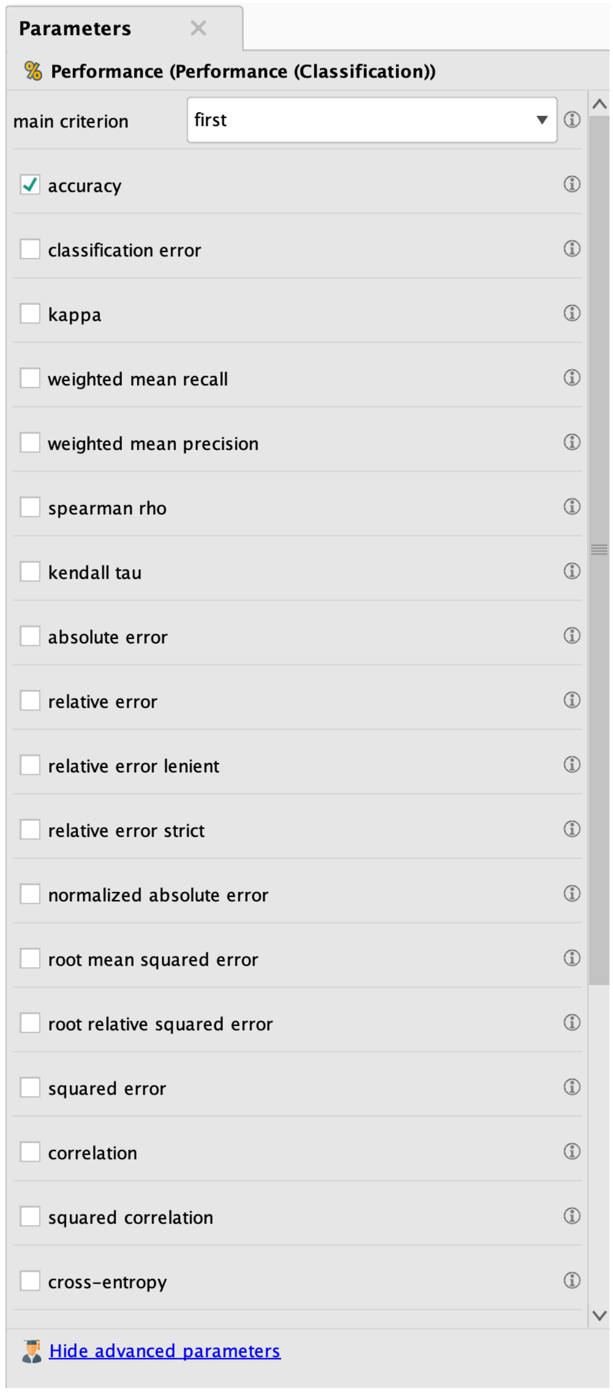
SET ROLES:





Set Roles: *Defines the purpose of the Diagnosis attribute as the label to predict*

PERFORMANCE:



Default: *Checks the accuracy of the model's predictions*

RESULTS:

Open in Turbo Prep Auto Model Interactive Analysis

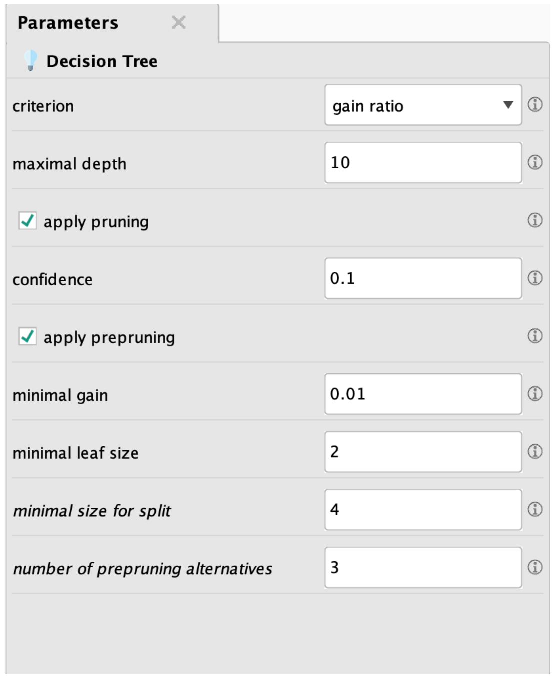
Filter (649 / 649 examples): all

Row No.	Diagnosis	Malaria_Prediction	Malaria_Risk_Chance	Age	Fever	Headache	Abdominal...	General_Bo...	Dizziness	Vomiting	Confusion	Ba
1	Negative	Negative	0	52	0	0	0	1	0	0	0	1
2	Positive	Positive	0.995	75	1	0	1	1	1	0	1	0
3	Positive	Positive	0.995	30	1	1	1	1	0	0	1	1
4	Positive	Positive	1	62	0	1	0	1	0	1	1	0
5	Positive	Positive	1	78	0	1	1	1	1	1	1	1
6	Positive	Positive	1	48	0	1	1	0	1	0	1	1
7	Negative	Negative	0	78	0	0	0	1	0	0	1	0
8	Positive	Positive	1	15	1	1	1	0	0	0	1	0
9	Positive	Positive	0.995	28	1	1	1	1	1	1	0	1
10	Negative	Negative	0	3	0	1	1	0	0	1	0	0
11	Positive	Positive	1	58	1	1	1	0	1	0	0	0
12	Positive	Negative	0.143	16	0	0	1	0	1	1	0	0
13	Positive	Positive	1	36	0	1	0	0	0	1	1	0
14	Positive	Positive	0.995	18	1	1	1	1	1	0	1	0
15	Positive	Positive	1	68	0	1	0	1	1	1	1	1
16	Positive	Positive	1	33	1	1	0	0	1	0	1	1
17	Positive	Positive	1	51	1	1	0	0	1	0	1	0
18	Positive	Positive	1	69	1	0	1	0	1	0	1	0
19	Positive	Positive	0.995	8	1	1	1	1	1	0	0	0
20	Positive	Positive	1	47	0	0	1	1	1	0	1	1
21	Positive	Positive	0.995	17	1	0	0	1	1	0	1	0
22	Positive	Positive	1	22	0	0	0	1	1	1	0	0
23	Positive	Positive	1	69	0	1	1	1	1	0	1	0
24	Positive	Positive	1	84	1	1	1	0	1	0	0	0
25	Positive	Positive	0.833	12	0	0	0	1	1	0	1	0
26	Positive	Positive	1	24	1	0	0	0	1	1	1	0

ExampleSet (649 examples,3 special attributes,12 regular attributes)

Results: *The table shows the output of our malaria prediction model, listing 649 patient cases with their predicted diagnosis, malaria risk and probability scores, and corresponding symptom data.*

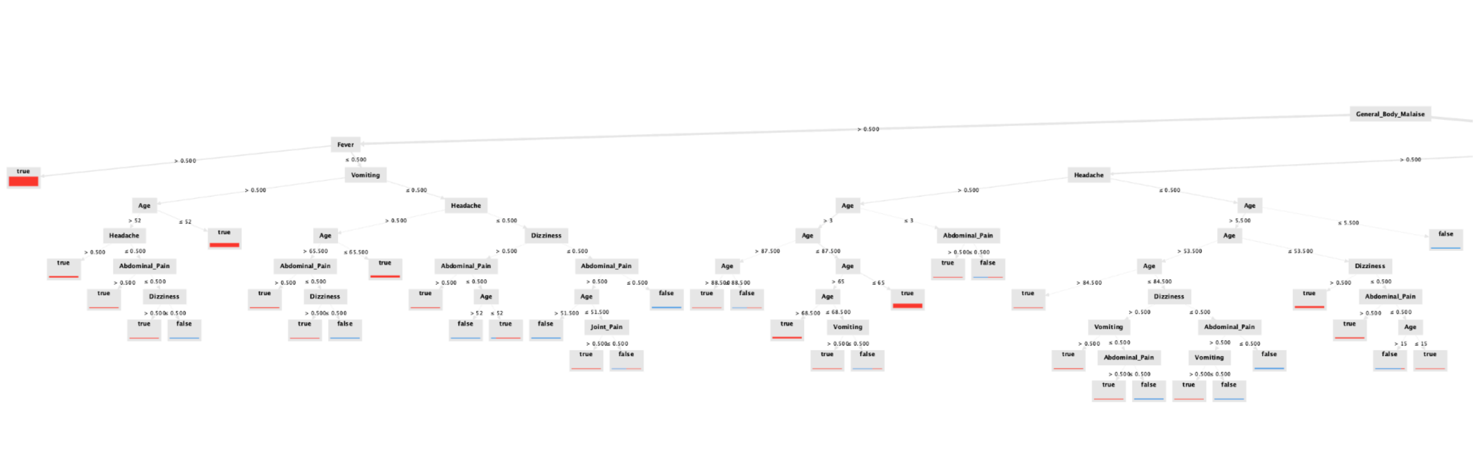
DECISION TREE PARAMETERS:



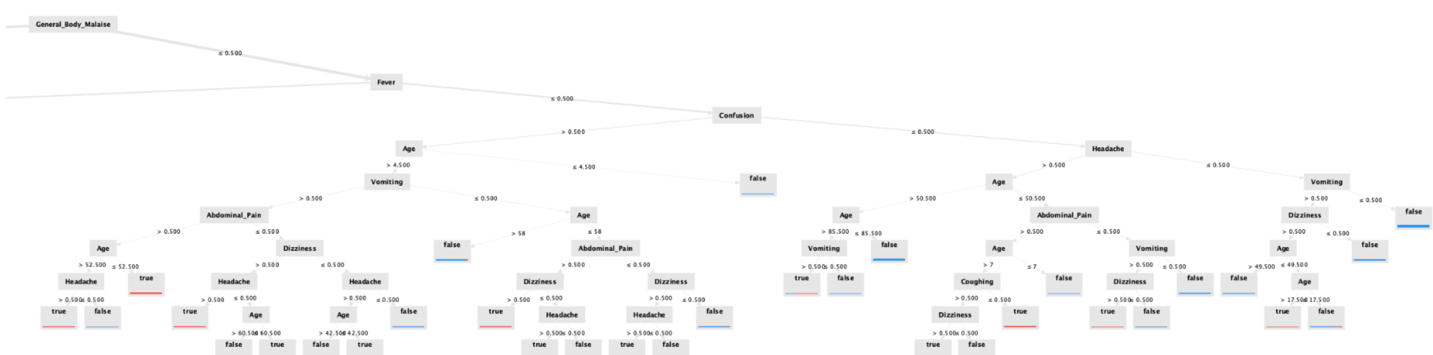
***DECISION TREE PARAMETERS**: Uses the gain ratio criterion with a maximum depth of 10 applies both pruning (confidence 0.1) and prepruning(minimum gain 0.01, minimum leaf size 2, minimum split size 4, and 3 prepruning alternatives) to control overfitting and improve model accuracy.*

DECISION TREE RESULTS:

Left-Side



Right-Side



***DECISION TREE**: Visually represents how the model classifies whether a person has malaria (True) or not (False) based on symptom patterns.*

ATTRIBUTE WEIGHTS:

attribute	weight
Dizziness	0.125
Headache	0.086
Abdominal_Pain	0.100
Vomiting	0.110
General_Body_Malaise	0.007
Joint_Pain	0.071
Confusion	0.056
Backache	0.109
Coughing	0.117
Fever	0.016
Chest_Pain	0.203

ATTRIBUTE WEIGHTS*: Shows the importance of each symptom or attribute in predicting malaria. Higher weights mean greater influence in the model's decision-making.*